SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0304	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	B Akshaya	**Reg. No.:**	192424336
Section / Batch:	Slot B/2024	**Date:**	10.08.2026

Code of Conduct

I B Akshaya(192424336)*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Akshaya

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

Ans:

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. MLE of Bigram Probability

Given:

- C(data) = 3
- C(data science) = 3

The MLE formula is:

P(science | data) = C(data science) / C(data)

= 3 / 3

= **1.0**

Answer:

P(science | data) = 1.0 or 100%

Interpretation:

Whenever the word **"data"** occurs in the given corpus, it is followed by **"science"**. Therefore, the keyboard can confidently predict **"science"** after the user types "data".

2. Backoff Model for "data science improves"

The sequence **"data science improves"** is unseen in the corpus.

The trigram:

P(improves | data, science) = 0

So, the model backs off to the bigram:

P(improves | science) = 0

Since this is also unseen, it finally backs off to the unigram:

P(improves) = 0

Therefore:

Backoff probability = 0

Interpretation:

Backoff is important because it allows the system to use lower-order information when a higher-order sequence is unseen. In a larger real-world corpus, this prevents the prediction system from completely failing for unseen sequences and provides better real-time predictions.

3. Deleted Interpolation for "data science is"

Given:

- $\lambda_1 = 0.5 \rightarrow$ Trigram
- $\lambda_2 = 0.3 \rightarrow$ Bigram
- $\lambda_3 = 0.2 \rightarrow$ Unigram

Formula:

P(is | data, science) = λ_1P(is | data, science) + λ_2P(is | science) + λ_3P(is)

From the given corpus:

P(is | data, science) = 2/3 ≈ 0.66

P(is | science) = 2/3 ≈ 0.66

For unigram:

Total words = 10

C(is) = 2

Therefore:

P(is) = 2/10 = 0.2

Now:

P = (0.5 × 0.66) + (0.3 × 0.66) + (0.2 × 0.2)

P = 0.33 + 0.198 + 0.04

P ≈ 0.568

Answer: P(is | data, science) ≈ 0.568

Interpretation:

Deleted interpolation combines trigram, bigram, and unigram information. This provides a balance between **accuracy from higher-order models** and **coverage from lower-order models**, especially when the training dataset is small or sparse.

4. Entropy Calculation

Given:

- P(is) = 0.66
- P(drives) = 0.33

Entropy formula:

$$H = -\Sigma P(x) \log_2 P(x)$$

Therefore:

$$H = -(0.66 \log_2 0.66 + 0.33 \log_2 0.33)$$

$$\approx 0.66(0.599) + 0.33(1.599)$$

$$\approx 0.395 + 0.528$$

$$\approx 0.923 \text{ bits}$$

Answer:

Entropy ≈ 0.92 bits

Interpretation:

The entropy is relatively low because **“is” has a much higher probability than “drives”**. Therefore, the keyboard has relatively low uncertainty when predicting the next word after “science”. Lower entropy means **higher prediction confidence**, while higher entropy indicates greater uncertainty.

Final Answers

Question	**Answer**
1. P(science | data)	**1.0**
2. Backoff P(improves)	**0**
3. Deleted Interpolation P(is | data, science)	**≈ 0.568**
4. Entropy	**≈ 0.92 bits**

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 2: AI-Powered Customer Support Chatbot

Aim

To implement a POS tagging system using the Penn Treebank POS Tagset and Hidden Markov Model (HMM) to identify the grammatical role of ambiguous words based on their context.

Procedure

1. Read the given user sentences.
2. Tokenize each sentence into individual words.
3. Assign appropriate Penn Treebank POS tags to every word.
4. Identify the grammatical role of the ambiguous word "book" using contextual clues.
5. Use the given HMM probabilities to calculate the likelihood of "book" being a Verb (VB) or Noun (NN).
6. Compare Rule-Based and Stochastic POS tagging approaches.
7. Analyze how POS tagging helps intent detection and response generation in chatbots.

1. POS Tagging of Given Sentences

Sentence 1

"Book a flight ticket now."

Word	POS Tag	Explanation
Book	VB	Verb – represents the action of booking
a	DT	Determiner
flight	NN	Noun

Word	POS Tag	Explanation
ticket	NN	Noun
now	RB	Adverb

Tagged sentence:
Book/VB a/DT flight/NN ticket/NN now/RB
Here, “book” is a Verb (VB) because it represents an action or command.
Sentence 2
“This book is interesting.”

Word	POS Tag	Explanation
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Tagged sentence:
This/DT book/NN is/VBZ interesting/JJ
Here, “book” is a Noun (NN) because it refers to a physical object.

2. HMM Probability Calculation
Given:
$P(book \mid VB) = 0.6$
$P(book \mid NN) = 0.4$
$P(Start \rightarrow VB) = 0.5$
$P(Start \rightarrow NN) = 0.5$
For Verb:
$P(VB, book) = P(Start \rightarrow VB) \times P(book \mid VB)$
$= 0.5 \times 0.6$
$= 0.30$
For Noun:
$P(NN, book) = P(Start \rightarrow NN) \times P(book \mid NN)$
$= 0.5 \times 0.4$
$= 0.20$
Therefore:
VB = 0.30
NN = 0.20
Since 0.30 > 0.20, the HMM favors VB (Verb).
Thus, in “Book a flight ticket now”, the word “book” is more likely to be tagged as a Verb.

3. Rule-Based vs Stochastic Tagging
Rule-Based Tagging
Rule-based tagging uses predefined grammatical rules and dictionaries.
Advantages:
- Simple to understand.
- Easy to implement.
- Works well for known grammatical patterns.

Limitation:
- Difficult to handle large vocabulary and ambiguous words.

Stochastic (HMM) Tagging
HMM tagging uses probabilities learned from training data.
Advantages:
- Handles lexical ambiguity using context.
- Suitable for large datasets.
- Can adapt better to different word usages.

Recommendation:

For a large-scale chatbot, Stochastic/HMM tagging is more suitable because it uses probabilities and contextual information to resolve ambiguous words such as “book”.

4. Role of POS Tagsets and Word Classes

Standardized POS tagsets such as the Penn Treebank Tagset provide consistent grammatical labels for words.

They help the chatbot to:

- Identify the grammatical structure of user messages.
- Improve intent detection.
- Understand whether a word represents an action or an object.
- Generate more appropriate responses.
- Improve overall language understanding and chatbot performance.

For example, identifying “book” as VB in *“Book a flight ticket now”* helps the chatbot recognize a flight-booking intent, while identifying “book” as NN in *“This book is interesting”* indicates that the user is talking about a book.

Result

Thus, the POS tags were successfully assigned using the Penn Treebank Tagset. The HMM calculation showed that “book” is more likely to be a Verb (VB) in the first sentence because its probability is 0.30, compared with 0.20 for Noun (NN). POS tagging and probabilistic models improve contextual understanding and intent detection in AI-powered chatbots.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

CASE STUDY 3: News Analytics and POS Tag Correction System

Aim

To apply Transformation-Based Tagging using a learned Brill Tagger rule to correct POS tagging errors, analyze word frequency distribution, and understand uncertainty in POS tag assignments using entropy.

Procedure

1. Take the given sentence and its initial POS tags.
2. Apply the learned transformation rule to the initial tags.
3. Identify and explain the incorrect POS tag using grammatical structure and sentence meaning.
4. Calculate the total word frequency and frequency distribution from the given corpus.
5. Explain how word frequency supports probabilistic POS tagging.
6. Compare the initial and corrected POS tags.
7. Discuss how entropy can be used to measure uncertainty before and after transformation.

1. Apply the Transformation Rule
Given sentence:
"Economic growth increases employment."
Initial tags:
economic/JJ
growth/NN
increases/NNS
employment/NN
Given rule:
Change **NNS to VBZ** if the preceding word is tagged as **NN**.
The word **"increases"** is tagged as **NNS**, and the preceding word **"growth"** is tagged as **NN**.
Therefore, the rule is applied:
increases/NNS → increases/VBZ
Corrected POS Sequence
economic/JJ
growth/NN
increases/VBZ
employment/NN
The correction is linguistically appropriate because **"increases"** functions as a third-person singular present-tense verb in the sentence. The sentence follows the structure:
Subject + Verb + Object
Economic growth → increases → employment
Subject Verb Object

2. Analysis of Initial Tagging Errors
The initial tag **NNS** for **"increases"** is incorrect because NNS represents a plural noun.
In this sentence, **"growth"** is the subject and is singular. Therefore, the word **"increases"** acts as the main verb agreeing with the singular subject.
Correct tags are:

Word	**Initial Tag**	**Corrected Tag**	**Reason**
economic	JJ	JJ	Describes growth
growth	NN	NN	Singular noun and subject
increases	NNS	VBZ	Third-person singular verb
employment	NN	NN	Noun functioning as object

Thus, the transformation rule correctly changes **NNS → VBZ**.

3. Word Frequency Distribution
Given corpus frequencies:

Word	**Frequency**
economic	120
growth	450
increases	210
employment	380

Total Frequency
120 + 450 + 210 + 380 = 1160
Total number of occurrences = **1160**
Frequency Distribution
Formula:
P(word) = Frequency of word / Total frequency

Word	**Frequency**	**Probability**
economic	120	0.1034

Word	Frequency	Probability
growth	450	0.3879
increases	210	0.1810
employment	380	0.3276

Therefore, **growth** has the highest frequency, followed by **employment**, **increases**, and **economic**.

Frequency information helps probabilistic POS taggers estimate how likely a word or word-tag combination is. However, frequency alone is not enough to resolve ambiguity; contextual information is also important.

4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging

Initially:

economic/JJ growth/NN increases/NNS employment/NN

After applying the learned rule:

economic/JJ growth/NN increases/VBZ employment/NN

The transformation-based tagger improves accuracy by identifying an incorrect tag and applying a context-based correction rule.

Entropy

Entropy measures the uncertainty associated with possible tag assignments.

A high entropy value indicates **high uncertainty**, while a low entropy value indicates **greater confidence** in the tag assignment.

Before applying the rule, the word **"increases"** may have multiple possible interpretations such as NNS or VBZ, resulting in greater uncertainty.

After applying the rule:

growth/NN → increases/VBZ

the contextual rule strongly favors **VBZ**, reducing uncertainty.

Therefore:

Before transformation → Higher uncertainty

After transformation → Lower uncertainty

The exact numerical entropy cannot be calculated from the given information because probabilities for the alternative POS tags are not provided.

Result

Thus, the Transformation-Based Tagger successfully corrected **increases/NNS** to **increases/VBZ** using the learned contextual rule. The frequency analysis showed that **growth** has the highest frequency of 450 occurrences. Transformation-based correction improves POS tagging accuracy and can reduce uncertainty in ambiguous tag assignments.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________